

1. Metric Selection & Interpretation

- **Why Accuracy is Insufficient:** In real-world AI/ML systems, accuracy alone is misleading because class imbalance is extremely common. If a dataset has 99% benign cases and 1% malignant cases, a model could simply guess "benign" every time and achieve 99% accuracy, while completely failing to identify the actual disease.
- **The Confusion Matrix:** This provides a detailed error analysis. It breaks down predictions into True Positives, False Positives, True Negatives, and False Negatives.
- **Precision vs. Recall:** Precision and Recall are business-critical metrics. In medical diagnosis, predicting a False Negative (telling a patient they are healthy when they actually have a malignant tumor) is incredibly dangerous. Therefore, if recall is low, the model is failing to identify the positive cases effectively.
- **F1-Score:** The F1-Score provides a balance between precision and recall. It is the preferred metric for imbalanced data because it accounts for both False Positives and False Negatives, unlike standard accuracy.
- **ROC Curve and AUC:** The ROC Curve measures the model's discrimination ability[cite: 4]. A higher AUC score indicates the model is highly capable of distinguishing between malignant and benign classes[cite: 4].

2. Handling Class Imbalance

- **The Challenge:** Class imbalance is a real-world data challenge where one target class heavily outnumbers the other[cite: 4].
- **Technique Applied (Stratification):** During the train-test split, the stratify=y parameter was used to preserve the original class distribution in both the training and testing sets[cite: 4].
- **Technique Applied (Class Weights):** To further handle class imbalance conceptually and practically, a second Logistic Regression model was trained using the class_weight="balanced" parameter[cite: 4]. This penalizes the model more for making mistakes on the minority class, forcing it to pay equal attention to both outcomes[cite: 4].

3. Final Model Decision & Comparison

- **Logistic Regression vs. Decision Tree:** The Logistic Regression model was compared against a Decision Tree Classifier[cite: 4].
- **Behavior and Stability:** Logistic Regression provides high stability and smooth probability curves (perfect for ROC analysis)[cite: 4]. Decision Trees, while offering high interpretability, are known for overfitting behavior[cite: 4].
- **Final Decision:** Because medical diagnoses require highly stable probabilities and reliable metrics, the Logistic Regression model (especially when balanced) is highly suitable for real deployment[cite: 4]. The Decision Tree, without hyperparameter constraints, is too prone to memorizing the training data, making the Logistic Regression the safer and more robust choice[cite: 4].